

DATA 2010 Project Report: An Empirical Study of Housing Price Drivers and Model Selection for Price Prediction

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General Guidelines

What you are doing: Pick a real dataset (Kaggle: <https://www.kaggle.com/datasets>), ask a few clear questions, clean the data, make good plots, apply techniques from this course, and write up what you learned.

What to submit:

- **Report PDF** (this document, please use this template for your report), submitted to **UM Learn**.
- **Code** as either:
 - a **public GitHub repo link**, or
 - a **ZIP file** containing your notebook(s) (.ipynb) / scripts.
- **Do not upload large datasets.** Include a link + instructions to download the data.

Target length: 4–6 pages (not counting appendix). Quality beats length.

Academic honesty (important):

- You **may** use starter code from class and standard library documentation.
- You **may** use Kaggle datasets, but your **analysis and writing must be your own**.
- If you borrow code/ideas, **cite the source** (link + one sentence).

How to get a strong grade:

- **Clear question:** not vague; you can answer it with data.

- **Real cleaning:** handle missing values, messy types, duplicates, outliers (as appropriate).
- **Good plots:** correct chart choice, labeled axes, readable, and they support your story.
- **Real analysis:** use multiple techniques from this course (not just one summary table).
- **Real-world insight:** explain what someone could do with your results.
- **Reproducible:** code runs; report is easy to follow.

1 Project Overview (At least Half Page)

Research question: What are you trying to find out?

Why it matters: Who would care about the answer, and why?

Your main result (write this last): summarize what you found.

2 Research Question (Make it Specific)

Write 3 questions you will answer. Good questions are measurable and not vague.

Examples of good project questions:

- “Which factors best predict X (e.g., house price, flight delay, customer churn)?”
- “How do outcomes differ across groups (e.g., regions, time periods, categories)?”

- “Can we cluster items/users into meaningful groups, and what characterizes each group?”

Example: Converted “\$1,234” to numeric by removing symbols/commas.

Bad questions (too vague): “What can we learn from this dataset?”

Example: Created a new feature `price_per_sqft`.

3 Data Source and Description

3.1 Where the data came from

- Dataset name:
- Link (Kaggle / open data portal / etc.):
- What each row represents:
- Key columns you used (2–8 columns):
- Any licensing/usage notes (if provided on the dataset page):

Example: Filled missing `age` with median within each group.

Show evidence: one small “before vs after” table or summary is great.

3.2 Basic dataset facts

Include a small table like this (fill with your values):

Number of rows	[#]
Number of columns	[#]
Time range (if any)	[e.g., 2018–2023]
Missing values present?	[Yes/No + brief note]

5 Exploratory Data Analysis and Visualization

You should include **at least 4** well-labeled figures. Every figure must answer: *What should the reader learn from this? Tell a story about your data and the plot.*

4 Data Cleaning and Feature Engineering

This is where you prove you handled real-world data.

Minimum expectations (pick what applies):

- Fixed data types (numbers stored as strings, dates, categories).
- Handled missing values (drop, impute, or justify).
- Removed duplicates / impossible values (if any).
- Dealt with outliers *carefully* (don’t delete without justification).

What you actually did: (bullet list)

5.1 Plot 1: [Title]

figure1.png

Figure 1: [What this figure shows. Mention the key takeaways.]

5.2 Plot 2: [Title]

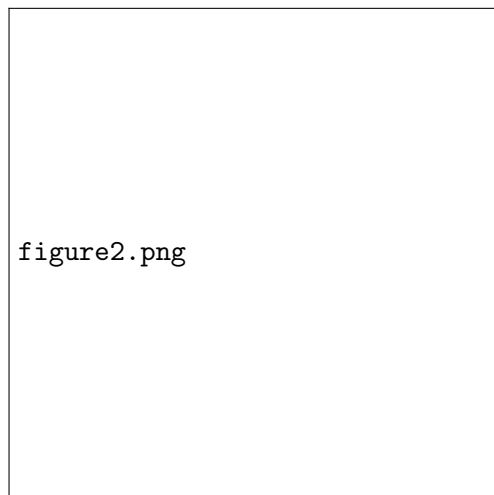


Figure 2: [Key takeaway.]

5.3 Plot 3: [Title]

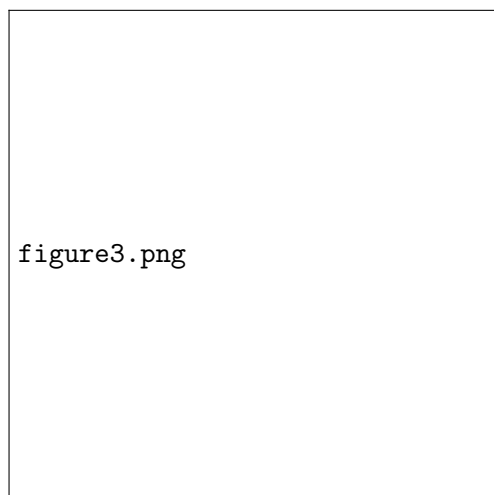


Figure 3: [Key takeaway.]

Common plot mistakes to avoid:

- Bar chart for a continuous variable (use histogram/density instead).
- No axis labels/units.
- Too many categories (group small ones into “Other” or filter).

6 Data Analysis (Techniques From This Course)

Pick a direction that fits your question. Use **multiple** techniques (not just one).

6.1 Analysis Plan

In 3–6 sentences: what methods did you use and why do they answer your question?

6.2 Baseline (If you do prediction)

If you predict something, include a simple baseline first (e.g., mean predictor, simple linear regression, majority class).

6.3 Main Results

Present results with at least **one table** and explain it in words.

Model / Method	Metric 1 (e.g., Accuracy)	Metric 2 (e.g., F1)
Baseline	[]	[]
Your method	[]	[]

If you used a train/test split: say how you split and why (and don’t evaluate on training data only).

6.4 Interpretation (So What?)

Don’t just report numbers. Explain:

- What the results mean in plain language,
- What surprised you (if anything),
- What limitations might change your conclusion.

7 Real-World Application

Your project should produce **useful insight** or a **decision** someone could act on.

Answer:

- Who is the stakeholder? (customer, city planner, student, business, etc.)
- What decision could they make based on your results?
- What would you recommend (1–3 bullets), and how confident are you?

8 Report Quality and Reproducibility

8.1 How to run your code

- Where your code is (GitHub/ZIP):
- Main file/notebook name:
- Steps to run (3–6 bullets):
- Required packages (if any):

8.2 Project structure (recommended)

- README.md (how to run)
- notebooks/ (analysis notebooks)
- src/ (helper functions, optional)
- figures/ (exported plots, optional)

9 Contributions (If Group Project)

List what each person did. Be honest and specific.

- Member 1:
- Member 2:
- Member 3:

10 Conclusion and Next Steps

Conclusion: 1 paragraph summarizing the answer to your question.

Next steps: If you had 1 more week, what would you improve or explore?

11 AI Disclosure

If you used AI tools (e.g., ChatGPT, Gemini) for coding or writing, disclose it.

- Tools used (if any):
- What you used them for (debugging, rewriting, brainstorming, etc.):
- Prompts you used:

References (No research papers required)

You do **not** need research papers. You should cite:

- Your dataset link (required),
- Any important documentation/tutorial you relied on,
- Any external code sources (if any).
- Dataset: [Name], [Link], accessed [date].
- Documentation: [Library/Tool], [Link].

A Appendix (Optional)

Extra figures, extra tables, or small code snippets that would clutter the main report.